

Space Telescope Finds Clues of Giant Crashes Near Young Stars

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Long ago, the young Earth crashed into a Mars-sized object called Theia. Scientists think pieces from that crash came together to form our Moon.

Now, NASA's James Webb Space **Telescope** is helping scientists look for similar crashes around young stars. A team led by Kate Su studied 21 rare **disks** of warm dust close to young stars. Only about 1% of young stars show this kind of dust.

The scientists studied the **minerals** in the dust. About one-third of the **disks** have lots of **silica**, a material found in a volcanic glass called **obsidian**. The team thinks these came from very powerful crashes between Mars-sized worlds. The other **disks** seem to come from gentler bumps, including between Moon-sized objects.

"This is the first time we have gathered enough systems to truly understand this subclass that we call extreme debris **disks**," Su said. The study was published on October 1.

Word Treasure

Telescope -- A tool that makes faraway things in space look closer and clearer.

disks -- Flat, round shapes, like rings of dust around a star.

minerals -- Natural solid materials that make up rocks.

silica -- A material found in sand, glass and some rocks.

obsidian -- A shiny black glass made when lava cools very fast.

Background you need

Planets are built when bits of rock and dust around a young star bump and stick together. Space telescopes like Webb help scientists see the dust left over from these crashes.

Break it down

WHO: A team of scientists led by Kate Su, using NASA's James Webb Space Telescope

WHAT: They found signs of giant crashes between big worlds in dusty disks

WHERE: Around young stars far out in space

WHY: To learn how planets like Earth, and our Moon, may have formed

Why it matters

Learning about crashes around other stars helps us understand how our own Earth and Moon were made.

Test what you remember

1. What object do scientists think crashed into the young Earth?

- ☐ (A) The Sun
- ☐ (B) Theia
- ☐ (C) Mars
- ☐ (D) Jupiter

2. What formed from pieces of that crash?

- ☐ (A) A comet
- ☐ (B) The Sun
- ☐ (C) Saturn's rings
- ☐ (D) Our Moon

3. Which telescope did the scientists use?

- ☐ (A) A telescope on the Moon
- ☐ (B) A backyard telescope
- ☐ (C) The James Webb Space Telescope
- ☐ (D) A pair of binoculars

4. How many dusty disks did the team study?

- ☐ (A) 2
- ☐ (B) 1
- ☐ (C) 21
- ☐ (D) 100

5. What material was found in about one-third of the disks?

- ☐ (A) Water
- ☐ (B) Ice cream
- ☐ (C) Gold
- ☐ (D) Silica

6. When was the study published?

- ☐ (A) June 26
- ☐ (B) January 2
- ☐ (C) October 1
- ☐ (D) December 17

Answer key (try the quiz first!): 1: B 2: D 3: C 4: C 5: D 6: C

MY THOUGHTS (at least 20 words):